Madhusree Mitra

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RESEARCH EXPERIENCE

Postdoctoral Researcher | Umeå Plant Science Centre
02/2021-04/2026 | Swedish University of Agricultural Sciences, Sweden

- standardized methods for qualitative and quantitative analysis of hydrophobic cell wall components [In prep]
- investigated roles of homogalacturonan and xylan modifications in cuticle and wood formation, contributing to improved growth and saccharification in Populus [1-5]
- analysed linkages between matrix polysaccharides and suberin -like lipids in the cell wall, characterised methodologies for these lipids [In prep]
- analysed drought induced changes in suberin-like lipids in the cell wall [In prep]
- analysed ferulic acid and p-coumaric acid content of rice straw using mass spectrometry (py-GC-MS, GC-MS and LC-MS).

UGC-Dr. D. S. Kothari Postdoctoral Fellow | Food Engineering and technology
06/2020 - 01/2021 | Institute of Chemical Technology, India

- utilised oleaginous yeast for biofuel production using brewery and bakery industry waste

Research Associate | Applied phycology and Biotechnology
04/2019- 03/2020 | Central Salt and Marine Chemicals Research Institute, India

- studied carbon sequestration in microalgal biomass and characterised high-value products

CSIR-GATE Junior and senior Research Fellow | Biotechnology and Phycology
09/2012 - 01/2019 | Central Salt and Marine Chemicals Research Institute

- engineered upstream and downstream processes to cultivate and isolate high-value microalgal products including omega-3 fatty acids, carotenoids, and phycobiliproteins. [6-8, 12]
- Characterised bioactive properties of C-phycocyanin and evaluated them as biosensors for heavy metals, and formulated omega-3 rich microalgal biomass for aquafeed production. [10-11]

EDUCATION

PhD. in Biological Sciences 2019 | Central Salt and Marine Chemicals Research Institute, India

B.Tech. in Biotechnology
2011 | West Bengal University of Technology, India

SKILLS

General:

- metabolomics &lipidomics • plant cell wall biology • mass spectrometry • chromatography • upstream & downstream bioprocess optimisation • microalgal cultivation • bioactivity & biochemical assays • analytical method development • large scale datasets/data processing

Analytical:

- Fluorescence, light & electron microscopy • UV-Vis/fluorescence/CD spectroscopy • Elemental analysis • Pyrolysis • NMR • FTIR

Data analysis:

- SIMCA • JMP • Agilent masshunter, profinder • Rstudio

Languages:

- Bengali: native
- English: fluent

RESEARCH PUBLICATIONS (Selected)

[1] Cuticle deposition in the leaf epidermal cells depends on integrity of homogalacturonan. iScience, 2025
[2] Wood-specific modification of glucuronoxylan can enhance growth in Populus. *Journal of Experimental Botany*, 2025
[3] Glucuronoyl Esterase Expressed in Aspen Xylem Affects γ-Ester Linkages Between Lignin and Glucuronoxylan Reducing Recalcitrance and Accelerating Growth. Plant Biotechnology Journal. 2025
[4] Modification of xylan in secondary walls alters cell wall biosynthesis and wood formation programs and improves saccharification. Plant Biotechnology Journal. 2025
[5] Xylan glucuronic acid side chains fix suberin-like aliphatic compounds to wood cell walls. New Phytologist. 2023
[6] A comparative analysis of different extraction solvent systems on the extractability of eicosapentaenoic acid from the marine eustigmatophyte *Nannochloropsis oceanica*. Algal Res, 2019
[7] Effect of glucose source on growth and fatty acid composition of a Euryhaline eustigmatophyte *Nannochloropsis oceanica* during mixotrophic cultivation. Bioresou Technol Rep, 2018
[8] Cultivation of *Nannochloropsis oceanica* biomass rich in eicosapentaenoic acid utilizing wastewater as nutrient resource. Bioresource Technology, 2016
[9] Effect of carbon supply mode on biomass and lipid in CSMCRI's *Chlorella variabilis* (ATCC 12198). Biomass and Bioenergy, 2016
[10] Fluorescence quenching property of C-phycocyanin from *Spirulina platensis* and its binding efficacy with viable cell components. Journal of Fluorescence, 2016
[11] C-Phycocyanin as a potential biosensor for heavy metals like Hg 2+ in aquatic systems. RSC Advances, 2016
[12] A euryhaline *Nannochloropsis gaditana* with potential for nutraceutical (EPA) and biodiesel production. Algal Research, 2015

COFERENCES (Selected)

- Metabolomics in Life Science, Umeå. 2024, 2026
- 3rd Nordic Metabolomics Conference, Trondheim, 2023
- Plant Cell wall Meeting, 2022, 2023, 2024

LEADERSHIP, MENTORING & OUTREACH

- Earned competitive research funding through the UGC-Dr. D. S. Kothari Postdoctoral scholarship and the CSIR-GATE PhD fellowship.
- Peer-reviewed scientific manuscripts for international journals within the fields of plant sciences and biotechnology.
- Served on the boards of the Umeå Postdoc Society, the SLU Postdoc Association, and the Swedish Network of Postdoc Associations.
- Coordinated and presented at public science communication events for *Pint of Science* Umeå and *Forskarfredag*.
- Mentored and co-mentored master's thesis students at the CSIR-Central Salt and Marine Chemicals Research Institute.